

2. Calculate the metrics afferent coupling, efferent coupling and instability for each class:

$AC(User) = 3$ (WebsiteMonitor, WebsiteMonitorController, main)

$EC(User) = 3$ (PreferredCommunicationChannel, Notification, WebsiteMonitorController)

$I(User) = 3/3+3 = 0,5$

$AC(WebsiteMonitorController) = 3$ (User, WebsiteMonitor, main)

$EC(WebsiteMonitorController) = 4$ (User, WebsiteMonitor, PreferredCommunicationChannel, Notification)

$I(WebsiteMonitorController) = 4 / 4+3 = \sim 0,6$

$AC(WebsiteMonitor) = 1$ (WebsiteMonitorController)

$EC(WebsiteMonitor) = 5$ (Website, WebsiteMonitorController, User, Notification, PreferredCommunicationChannel)

$I(WebsiteMonitor) = 5 / 5+1 = \sim 0,85$

$AC(Website) = 1$ (WebsiteMonitor)

$EC(WebsiteMonitor) = 0$

$I(WebsiteMonitor) = 0 / 0+1 = 0$

$AC(PreferredCommunicationChannel) = 3$ (WebsiteMonitorController, WebsiteMonitor, User)

$EC(WebsiteMonitor) = 0$

$I(WebsiteMonitor) = 0 / 0+3 = 0$

3. Suggest a package structure for your implementation:

Used Structure of my Implementation:

main.java

com.uas.WebsiteMonitor

--monitor

----WebsiteMonitor.java

--Controller

----WebisteMonitorController.java

--User

----user.java

--util

----Notification.java

----PreferredCommunicationChannel.java

4. Commit your software to a new github.com repository:

→ See: <https://github.com/lilsemy/SWED> (folder Ex4)

5. Name options to reduce coupling between your packages:

I could reduce the coupling of my packages by introducing interfaces, over which the single modules can communicate. E.g. WebsiteMonitor does not need as an attribute the whole instance of a user but could also just use an Interface of User, from which it only gets the ID or something like this. Also the Observer Pattern could help to fix the problem, that e.g. the WebisteMonitorController needs for nearly every function the whole user instance. That could also just be done, by an Observer instance in between, that delivers the view of the User instance, without the need, that the Controller needs the whole instance.